

ADAMA SCIENCE AND TECHNOLOGY UNIVERSITY

SCHOOL : Applied Natural Science
Subject : Phys 1101, General Physics
Total Marks : 20

PROGRAM: Applied Physics
Assessment : Mid
Time : 2 hours

Name : _____

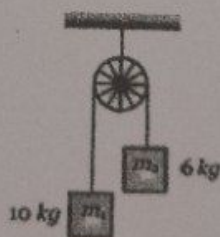
Id. No. _____

General Instruction

- Check the availability of all the questions.
- No lending and borrowing inside the exam hall.
- Mobile phone, written manuscripts and printed materials are strictly prohibited inside the exam hall and that will be considered as a guilty of malpractice.
- Must give a neat diagram and units wherever it is necessary. Give the answer based on the mark distribution.

Part – A Short answer questions ($8 \times 1.5 = 12$ marks)

1. If a car covers $2/5^{\text{th}}$ of the total distance with v_1 speed and $3/5^{\text{th}}$ distance with v_2 then average speed is
2. A projectile is fired at 30° to the horizontal. The vertical component of its velocity is 80 ms^{-1} . Its time of flight is T . What will be the velocity of the projectile at $t = T/2$?
3. A projectile is projected with initial velocity $(6i + 8j) \text{ m/s}$. If $g = 10 \text{ ms}^{-2}$, then horizontal range is
4. Find the magnitude and direction cosines of the vectors $3i + 7j - 4k$, $i - 5j - 8k$ and $6i - 2j + 12k$.
5. Two masses m_1 and m_2 are attached to a string which passes over a frictionless smooth pulley. When $m_1 = 10 \text{ kg}$, $m_2 = 6 \text{ kg}$, the acceleration of masses is



6. A body of mass 5 kg rests on a rough horizontal surface of coefficient of friction 0.2 . The body is pulled through a distance of 10 m by a horizontal force of 25 N . The kinetic energy acquired by it is ($g = 10 \text{ ms}^{-2}$)
7. Explain stress strain diagram.

8. The water flowing from a garden house fills a 15 litre container in one minute. Find the speed of water from the un obstructed opening of radius 1cm and the velocity from the obstructed opening whose area is half of the previous one.

Part - B Long answer questions (8 marks)

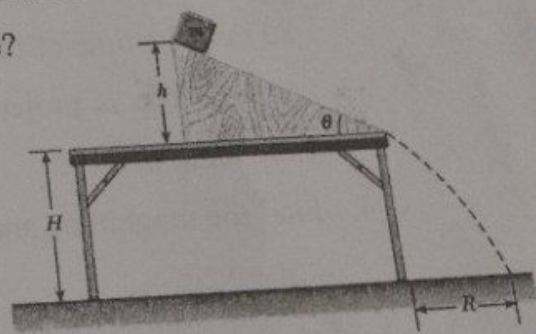
1. A projectile is fired with an initial velocity u making an angle θ with the horizontal. Then

- Show that its trajectory is a parabola
- find the expression for time of flight ,
- find the expression for horizontal range, and
- find the expression for the maximum height.

- 4 marks

2. A block of mass $m = 3\text{kg}$ is released from rest at the height of $h = 0.5\text{m}$ above the surface of a table, at the top of a $\theta = 37^\circ$ incline as shown in figure. The coefficient of kinetic friction of incline is $\mu = 0.3$ fixed on a table of height $H = 2\text{m}$.

- Determine the acceleration of the block as it slides down the incline
- What is the velocity of the block as it leaves the incline?
- How far from the table will the block hit the floor?
- How much time has elapsed between when the block is released and when it hits the floor?



- 4 marks